Applied University
Faculty of Engineering Technology
Basic Science Department
Department of Physics and Basic Sciences

جامعة العلوم التطبيقية
كلية الهندسة التكنولوجية
قسم الفيزياء والعلوم الأساسية

Time of exam (90 minute) Exam Phy-111(B)

2-5	(11-2)

الاسم بالعربية : أحمد اسامة SPU.fet مدرس المادة

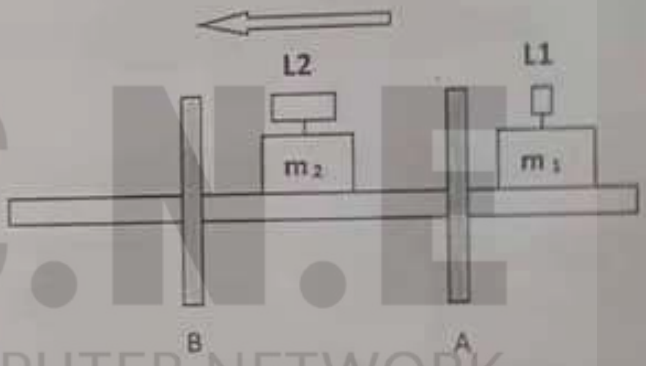
Note: When you need consider the acceleration of gravity (g) =10 m/s²

Q:1	1	2	3	4	5	6	7	8	9	10	11	12
	C	A	A	B	A	A	D	D	D	A	D	A

Q:2	13	14	15	16	17	18	19	20	21	22	23	24	25
	(T) F	T (F)	(T) F	(T) F	(T) F	(T) F	T (F)	T (F)	(T) F	T (F)	T (F)	T (F)	(T) F

Q.1: Circle the correct answers, and then move these answers to the table :(24 marks)

In collisions experiment, according to the figure Shown, answer (Q1), (Q2), (Q3)



1) In inelastic collision experiment, m_1 collides m_2 which In static state .If m_1=200g, m_2=100g, L_1=3cm, and A_1=0.3 s. The linear momentum for m_1 and m_2 after collision (P_f) is: (Where (A and B) are photo gates)

A) 0.01 kg.m/s	B)0.02 kg.m/s	C)0.03 kg.m/s	D) 0.012 kg.m/s

2) In the previous collision, we can say:

A) $P_i = P_f$	B) $K_i \neq K_f$	C) $V_{1f}=V_{2f}$	D) all of them

3) If (B_2=0.06 s),(B_1 =0.12 s). The length of the flag (L_2) ,which positioned on (m_2) is :

A) 6 cm	B) 8 cm	C) 4 cm	D) 5 cm

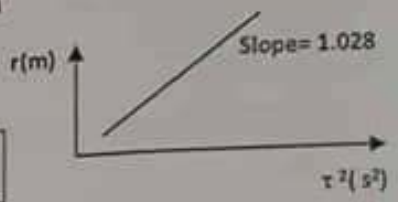
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4) In Hooks law experiment (m) oscillates under the effect of spring, which has the spring constant (k), if the mass(m) Increased to (9m), the period of one oscillation (τ) will be:

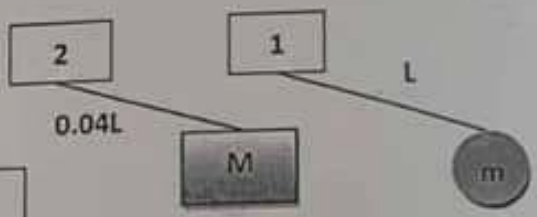
A) 0.3 τ	B) 3 τ	C) 2 τ	D) 0.2 τ

5) In uniform circular motion, if (m=10g),(M=45g),then the acceleration Of gravity (g) is:



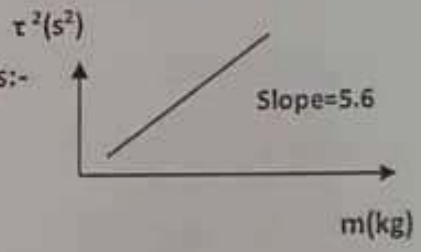
A) 9 m/s^2	B) 10 m/s^2	C) 8 m/s^2	D) 11 m/s^2

6) If the periodic time for pendulum (1) is ($\tau_{(m)}$)=0.11s) then (t_{20}) for The pendulum (2):



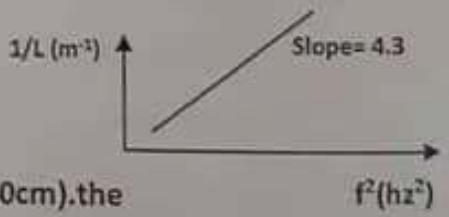
A) 0.44 s	B) 0.32 s	C) 0.6 s	D) 0.48 s

7) In Hooks law experiment, depending on graph the spring constant (K) is:-



A) 6 N/m	B) 8 N/m	C) 9 N/m	D) 7 N/m

8) In simple pendulum experiment, we got this graph shown. The acceleration of gravity (g) is :



A) 10.4 m/s^2	B) 11 m/s^2	C) 8 m/s^2	D) 9.2 m/s^2

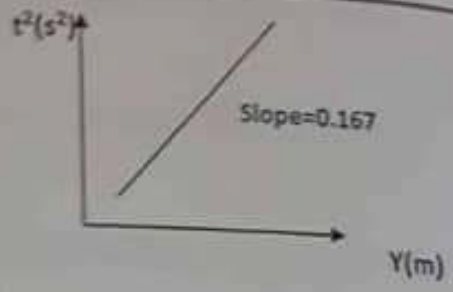
9) In uniform circular motion experiment, (m=10g), (M=80g) and (r=20cm).the Centripetal acceleration (a_c) is:

A)30 m/s^2	B)40m/s^2	C)50m/s^2	D)80m/s^2

10) In Newton's second law experiment when (Mw=0.15 kg) , (mh=15g) .the acceleration of (Mw) is

A)0.9 m/s^2	B) 1.3 m/s^2	C)1.4 m/s^2	D)1.2 m/s^2

11) In free falling experiment, student got the graph shown



The acceleration of gravity is:

A)8 m/s^2	B) 9 m/s^2	C)11 m/s^2	D)12 m/s^2

12) In the previous question, the percentage error in acceleration of gravity,

As the true value of ($g=9.8\ m/s^2$) :

A) 22.4 %	B) 12.2 %	C) 18.37 %	D) 8.89 %

Q (2): Put (T) in front of the correct phrase, and (F) In front of the wrong phrase. Then move these answers to the table. (26marks)

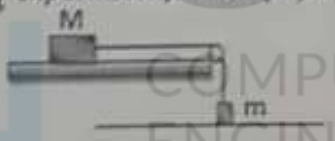
13) The instrument shown called photo gate ( ).



14) The instrument shown called air table ( ).



15) In gravitational potential energy experiment, when (m) reach the ground, the (M) will move in constant speed.( )



16) The benefit of crocodile clip in uniform circular motion is to keep the the magnitude of velocity constant in the experiment. ( ).



17) In gravitational potential experiment, the relation between (hi) and time (t) when (M) and (m) were constant in the experiment is inversely proportional ( ).

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18) N/m is the unit of spring constant. ( ).

19) The number (1.000250) contains (6) significant figures. ( ).

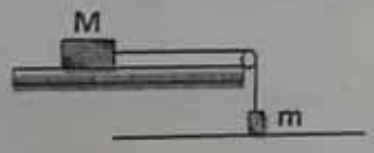
20) If the time of (30) oscillation is (6.0s), then the frequency is (5.4 Hz). ( ).

21) In all kind of collisions always ($P_i = P_f$).( ).

22) If A=5.89mm, B=2.9mm.as(C =A – B). Then (C±△C) =(2.99 ± 0.06)mm. ( ).

((using significant figures rules))

23) In gravitational potential energy experiment (△U) for (m) and (M).( ).



24) If $y(t)=3t^2-2t+4$.the Instantaneous velocity at t=2s, v(2)=12 m/s.( ).

25) In gravitational potential energy experiment, as(h_i=80cm) ,(M=0.190 kg) and(m_h=10 g),then the velocity of (m_h) at height (h_f= 40 cm) ,is 0.63 m/s.( ).

GOOD LUCK TO ALL